

Electric Vehicles & Renewable Energy Production



State-level capstone analyzing whether cleaner electricity grids predict higher EV adoption using 2018 vehicle registrations and 1990–2018 generation data across 50 states + DC.

Executive Summary

Core Result

Renewable share shows a weak, non-significant correlation with EV adoption ($r = 0.24$, $p = 0.088$).

Policy vs Grid

Policy (incentives, ZEV mandates) better explains high EV adoption than grid cleanliness.

Scale & Scope

543,610 EVs registered (2018). National renewable share: 9% → 17% (1990→2018).

Objective & Guiding Questions

Test whether state renewable energy share predicts EV adoption, examine renewable growth, identify leading EV states, and assess the role of policy.



Compare

Renewable share vs EVs per 1,000 vehicles.



Trend

Track national and state renewable growth (1990–2018).



Policy

Assess incentives, ZEV mandates, and coalitions as explanatory factors.



Data & Methodology

Datasets

EV registrations (2018), total vehicle registrations (2018), energy generation by source (1990–2019), state reference codes.

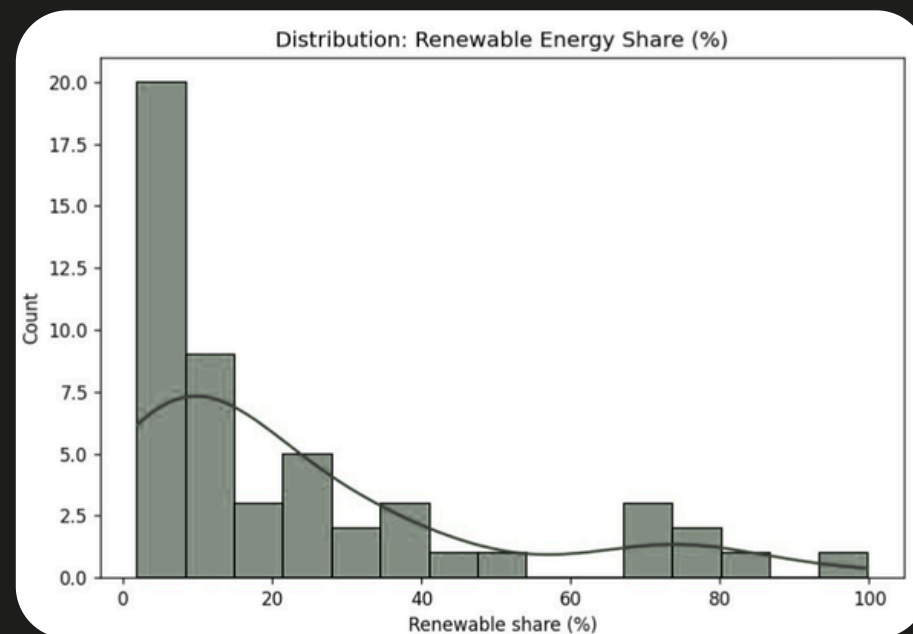
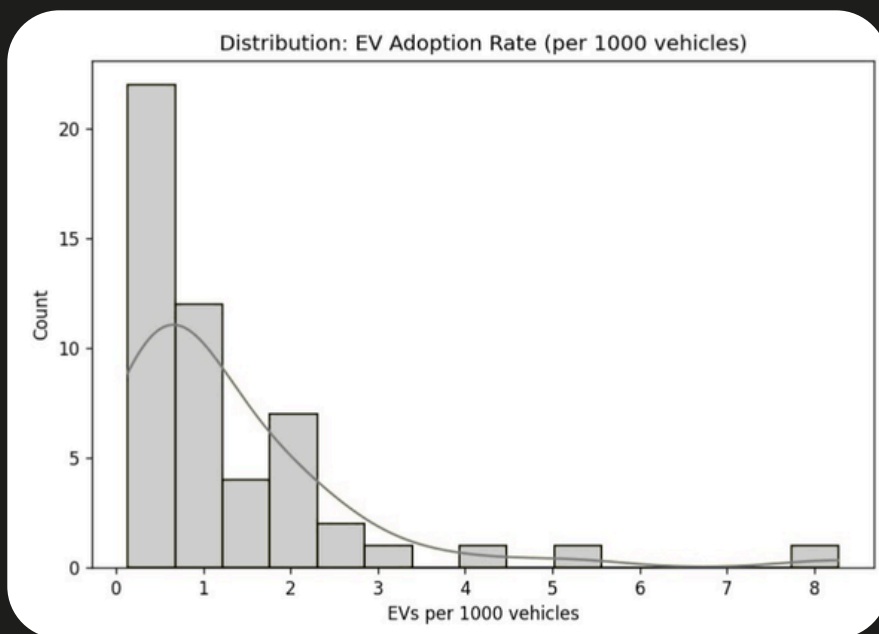
Cleaning

Standardized state names, filtered producer types, parsed malformed headers — final match: 51/51 with no nulls.

Primary Metric

EV adoption = EVs registered per 1,000 total vehicles (normalizes for state size).

Finding 1 – Concentrated EV Adoption

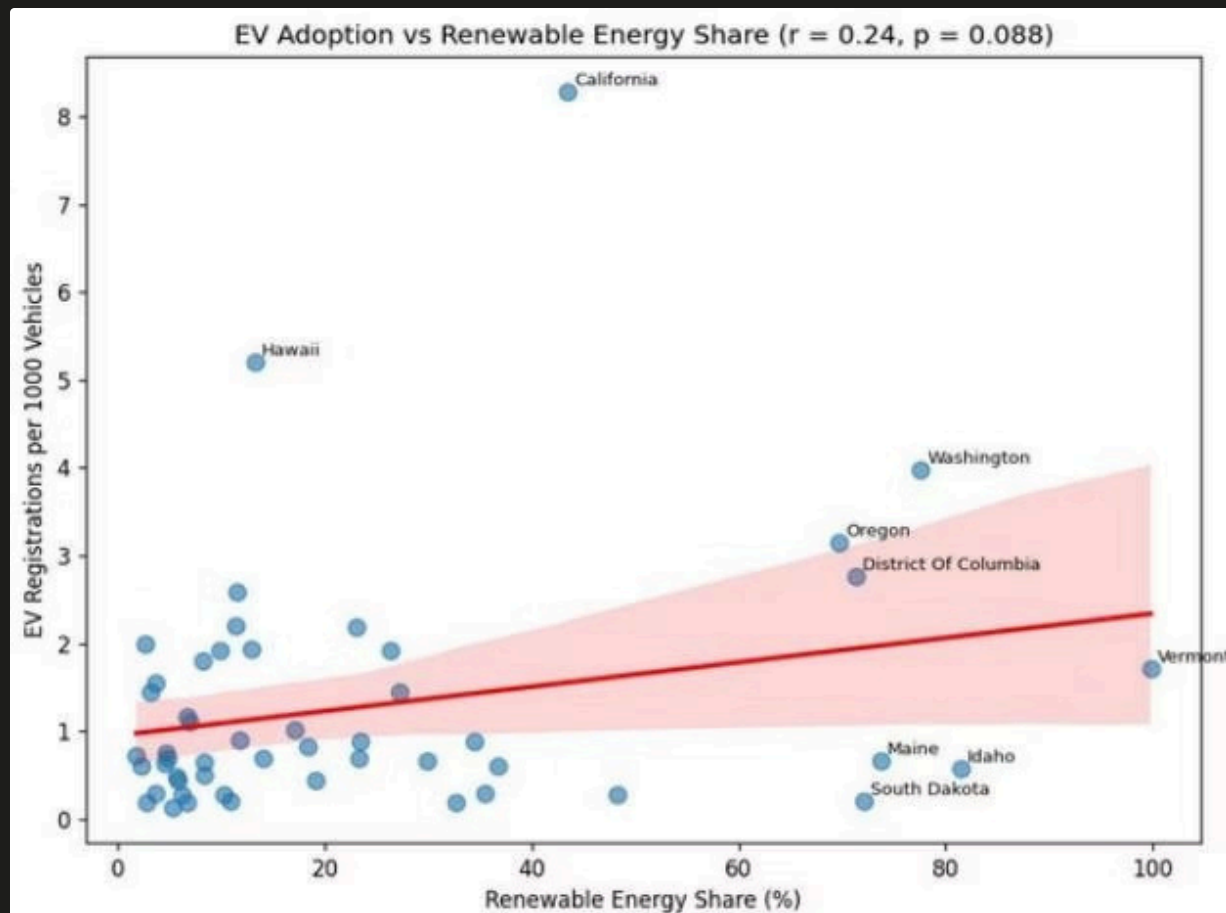


Distribution is right-skewed: median = 0.70 EVs per 1,000 vehicles. California leads at 8.28 (1.6× Hawaii, ~2.1× Washington). Most states cluster below 2 per 1,000.



EV adoption in 2018 was heavily concentrated in a few outlier states.

Finding 2 – Renewable Share vs EV Adoption

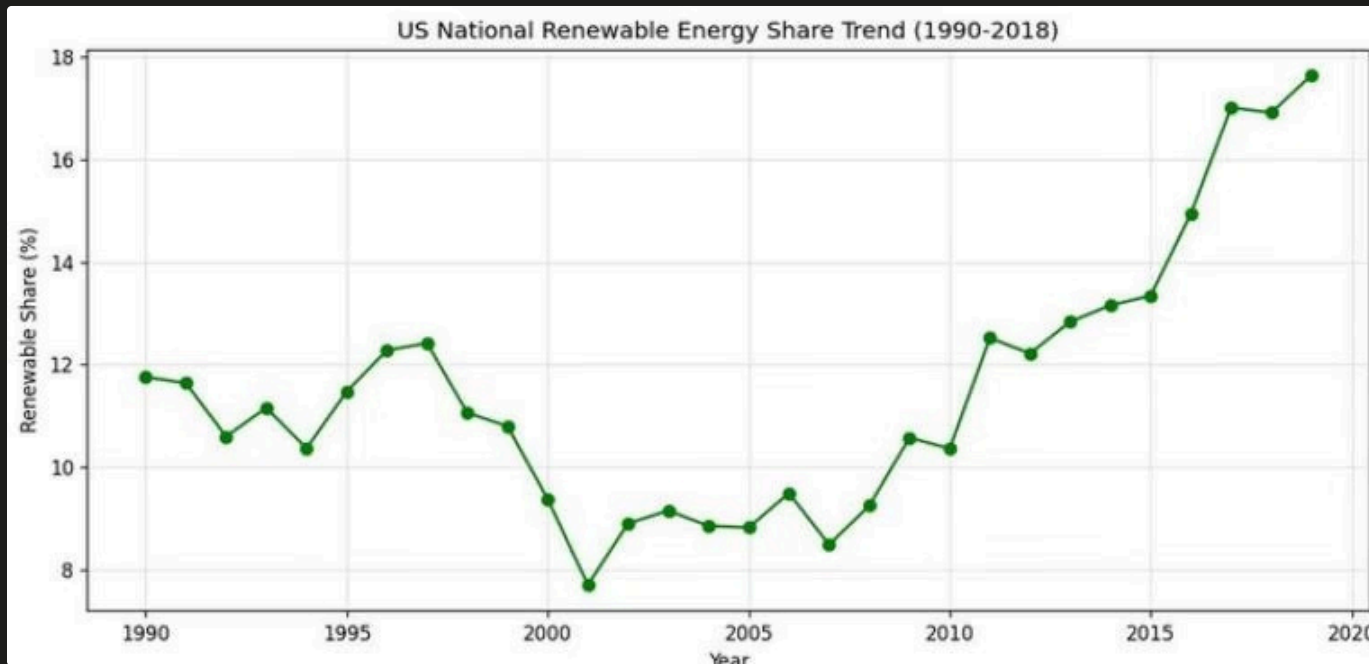


The scatter plot makes the disconnect visible: Vermont (99.8% renewable) and Idaho (81%) both show

low EV adoption, while California (43% renewable) and Hawaii (13% renewable) rank among the highest adopters. If clean electricity reliably drove EV adoption, these outliers would not exist in this pattern.

Scatter test: $r = 0.24$ (weak), $p = 0.088$ (not significant). Examples of disconnect: Vermont ~99.8% RE but low EVs; California ~43% RE with highest EV adoption.

Finding 3 – Renewables Growth: Midwest Wind Story



National renewable share rose from ~9% (1990) to ~17% (2018), with steep growth post-2015. Major growth drivers: Great Plains wind — Kansas, Iowa, Oklahoma jumped to ~35%+.

Those high-growth states do not overlap with top EV adopters, reinforcing independence of trends.

Finding 4 — Geographic Patterns



Maps show West Coast & Northeast concentrated EV adoption; high renewable shares often appear in Midwest/Great Plains. Exceptions (WA, OR) exist, but overlap is uncommon.

Policy Context — What Explains EV Adoption?



California

ZEV mandate + Clean Vehicle Rebate Project (direct rebates) — consistent high adoption.



Multi-state ZEV Coalition

States (CT, ME, MD, MA, NJ, NY, OR, RI, VT) coordinate mandates and incentives, boosting adoption independent of grid mix.



Colorado

State tax credit for EV purchases — an example of non-ZEV policy driving adoption.

Conclusion: coordinated regulatory and financial incentives, not clean grids, most consistently align with high EV adoption.

Conclusions & Recommendations

Key Findings

EV adoption concentrated;
renewable share not a reliable
predictor ($r=0.24$, $p=0.088$);
renewables growth led by Midwest
wind; policy aligns with EV leaders.

Recommendations

Policymakers: treat EV adoption and
grid decarbonization as separate
goals — use direct incentives, tax
credits, and ZEV mandates to spur
EV uptake.

For Researchers & Industry

Incorporate structured policy
datasets to quantify policy effects;
industry should prioritize markets
with active EV policy frameworks for
deployment and investment.

- ❏ Limitations: single-year EV data (2018) and qualitative policy assessment; correlation $\neq$ causation; other factors (density, chargers, manufacturers) unmeasured.



Thank you for checking this out

GITHUB : https://github.com/mynkbhardwaj87007-spec/ev_adoption_analysis

